(a) How one query reads image features

Illustrative sampling locations and rectangles

Deformable cross-attention

Native read in both detectors



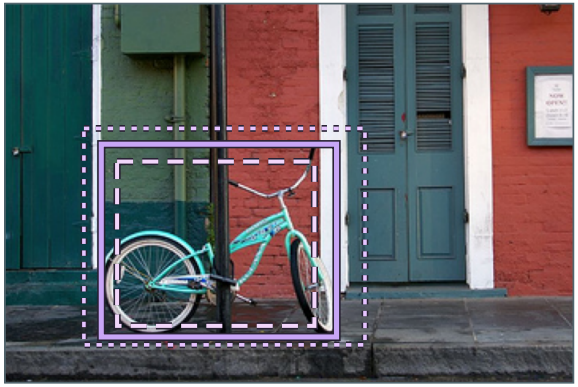
Sample points



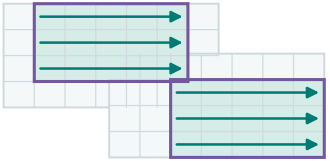
Weight the sampled feature values.

Boundary-conditioned state aggregation (BCSA)

Additional read in Boundary-State DETR



Rectangle states



Weight each rectangle state by its probability.

(b) The final decoder layer

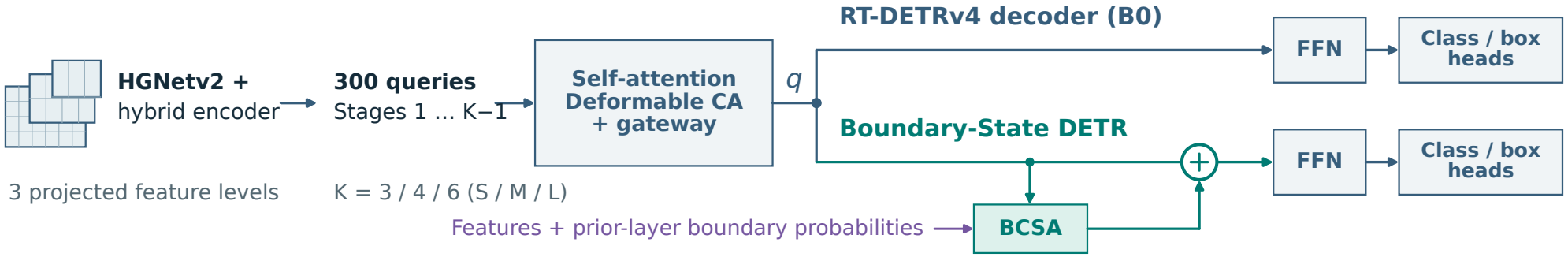
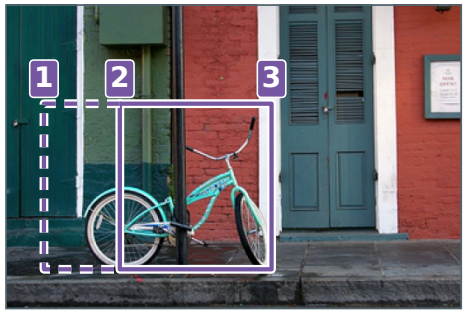


Fig. 1. Both detectors use deformable attention. In its final layer, Boundary-State DETR also weights rectangle states by the previous layer's boundary probabilities and adds the resulting correction to the query before the feed-forward network (FFN).

Blue Bike · Talbot Troy · CC BY 2.0 · COCO val2017 #426166 (cropped)

(a) Where BCSA reads

Two candidate boxes



Illustrative boundaries and states

Scan one feature row

Cell write	4	2	3	Prob.
Wide [1, 3]	0 → 4	→ 4	→ 5	0.25
Tight [2, 3]		0 → 2	→ 4	0.75

Reset at the left edge. Each cell updates $h_t = \frac{1}{2} h_{t-1} + u_t$.

Weighted read

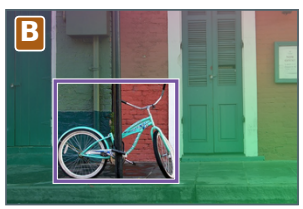
$0.25 \times 5 + 0.75 \times 4$

4.25

Expected state

(b) How BCSA trains with a second view





Original

$p^A,\ q^A,\ x^A,\ \beta^A$ →

Background edit

Only Δ_t^B →

BCSA Hybrid read with p^A and q^A

Retain prior state | Write new input

$$h_t = \exp(-\Lambda \Delta_t^B)\, h_{t-1} + \Delta_t^B\, (x_t^A \otimes \beta_t^A)$$

Training objective

$\mathcal{L}_{\text{native}}(A)$

$+ \lambda\, \mathcal{L}_{\text{det}}(\hat{y}_{\text{hybrid}}, y)$

Shared detection targets.

Inference

One view supplies all terms.

Fig. 2. Boundary-conditioned state aggregation (BCSA) weights each box's mean row state by its boundary probability. To train across backgrounds, we take the Mamba step size for both retention and writing from the edited view. Both predictions are supervised with the original class and box targets.

Photo: Talbot Troy · CC BY 2.0 · COCO val2017 #426166. Purple boxes in (b) mark the bicycle annotation.